SMART MANUFACTURING AI

Visual Quality Inspection & Automated Defect Intelligence Platform

RESULT: PASS

PASSED QUALITY AUDIT

Inspection ID:	INSP-1793369899-161E57	**Timestamp:**	2026-10-30 19:48:19
Component Name:	Tractor engine housing	**Similarity Score (SSIM):**	99.34%
Total Defect Regions:	0	**Affected Surface Area:**	0.000%
High-Severity Defects:	0	**Processing Duration:**	227.4 ms

Visual Evidence & Comparative Computer Vision Analysis

Figure A: Reference (Good Component)

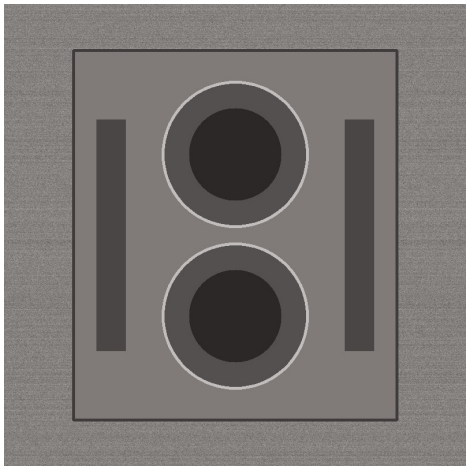


Figure B: Inspected Component

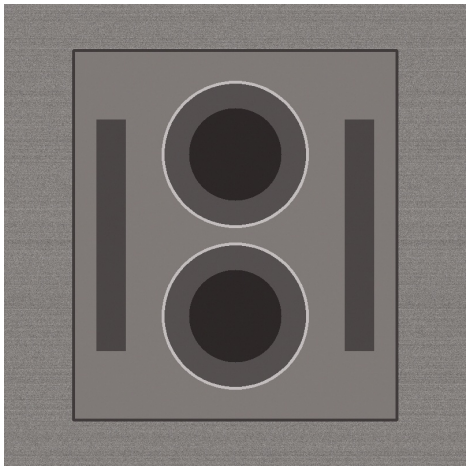


Figure C: SSIM Difference Heatmap

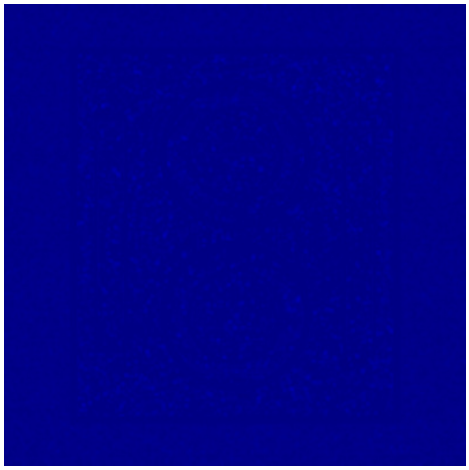
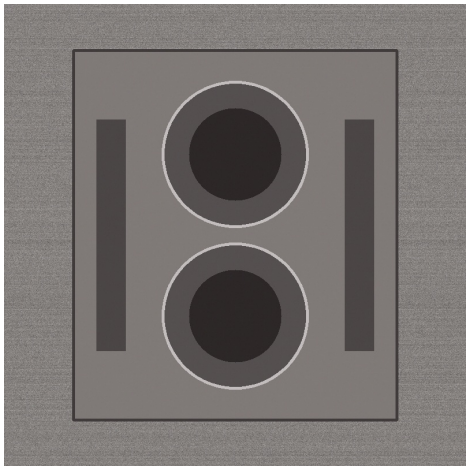


Figure D: Defect Highlighted & Annotated



Detected Defects & Anomaly Registry

No anomalous defects detected. Component complies with quality specification.

AUTOMATED QUALITY ENGINEERING RECOMMENDATIONS:

- Component conforms with engineering dimensional and cosmetic tolerances.
- Clear Tractor engine housing for downstream assembly line transport.
- Log component serial number to batch quality ledger.

Generated by Smart Manufacturing AI – Computer Vision Quality Inspection Intelligence Suite. Certified for plant floor auditor sign-off.